

Netflix

Content Analysis

*Data Cleaning, Exploratory Data Analysis,
and SQL Validation*

Exploring content distribution, audience
targeting, and growth trends across 8,797
Netflix titles

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Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn) |
PostgreSQL

Dataset: Netflix Movies and TV Shows Dataset

8,797

titles

6,131

Movies

2,666

TV Shows

01

SECTION 01

Project Overview

This report presents a comprehensive analysis of the Netflix Movies and TV Shows dataset. The objective was to clean, validate, and explore the dataset in order to uncover meaningful insights about content distribution, audience targeting, genre preferences, geographic representation, and platform growth.

The analysis was conducted using Python for data preparation, cleaning, visualization, and exploratory analysis, while PostgreSQL was used to independently validate key findings through structured SQL queries.

By examining over 8,700 Netflix titles, the project provides a data-driven view of how Netflix has developed its content catalogue and how its content strategy has evolved over time.

Dataset: Netflix Movies and TV Shows Dataset

Focus Areas:

- Content type distribution
- Geographic content representation
- Audience ratings and targeting
- Genre popularity
- Content release patterns
- Platform growth trends

SECTION 02

Data Familiarization and Assessment

The dataset was initially reviewed to understand its structure, identify key variables, and assess overall data quality.

The assessment focused on determining the number of records, reviewing data types, distinguishing between numerical and categorical variables, and identifying potential data quality issues that could affect analysis.

Particular attention was given to fields such as title, type, country, rating, release_year, and date_added, as these variables formed the foundation of the subsequent analysis.

The review revealed several missing values across multiple columns, highlighting the need for a structured data cleaning process before meaningful insights could be generated.

SECTION 03

Data Cleaning and Preparation

Before analysis, the Netflix dataset was cleaned and standardized to improve data quality and ensure reliable results.

3.1 Missing Value Treatment

Several columns contained missing values, particularly **director**, **cast**, **country**, **rating**, and **date_added**. The following actions were taken:

- Missing values in the **director**, **cast**, and **country** columns were replaced with **"Unknown"** to preserve records and avoid unnecessary data loss.
- Missing values in the **rating** column were filled using the most common rating category.
- Records with missing **date_added** values were removed because they could not support time-based analysis.

3.2 Duplicate Check

The dataset was checked for duplicate records to ensure data integrity.

- Duplicate rows were identified and reviewed.
- Any exact duplicate records were removed to prevent double-counting during analysis.

3.3 Date Formatting

The **date_added** column was converted from text format to a proper datetime format.

This enabled:

- Year-based analysis
- Content growth analysis
- Content addition trend analysis

3.4 Column Standardization

Column names were reviewed and standardized for consistency and ease of use during analysis.

Examples include:

- Using lowercase column names
- Maintaining underscore naming conventions

This improved readability and compatibility across both Python and PostgreSQL.

3.5 Feature Engineering

A new field called **year_added** was derived from the **date_added** column.

This field was created to support:

- Analysis of titles added per year
- Netflix growth trend analysis
- Content acquisition pattern analysis

3.6 Data Validation

Several validation checks were performed to confirm dataset quality:

- Missing values were rechecked after cleaning.
- Duplicate records were verified and removed.
- Date conversions were validated.
- Content ratings and country values were reviewed for consistency.
- The **show_id** field was confirmed as the unique identifier for each title.

3.7 Clean Dataset Output

After completing the cleaning process, the final dataset contained **8,797 records** and was exported as **netflix_cleaned.csv** for exploratory analysis and PostgreSQL validation.

The cleaned dataset provided a reliable foundation for generating insights and visualizations throughout the project.

SECTION 04

Exploratory Data Analysis (EDA)

After data cleaning and validation, exploratory analysis was conducted to understand the composition of Netflix's content library, identify audience trends, evaluate geographic representation, and examine the platform's growth over time.

4.1 KPI Summary

The following headline metrics were derived from the cleaned dataset:

8,797	6,131	2,666	100+	1925
Total titles	Movies	TV Shows	Countries	Oldest title

4.2 Content Type Distribution

The first stage of the analysis examined the balance between Movies and TV Shows within the Netflix catalogue.

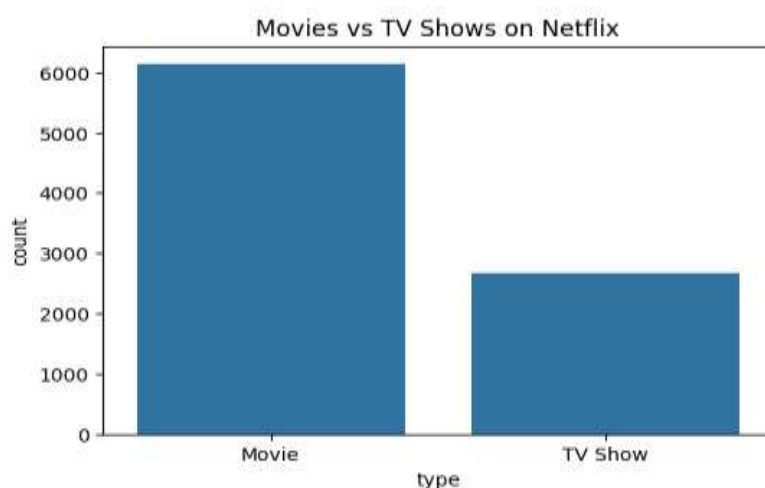


Figure 1: Movies vs TV Shows Distribution

Key observations:

- Movies account for approximately 70% of all titles on Netflix.
- TV Shows represent the remaining 30% of the catalogue.
- The findings suggest that film content continues to be the dominant component of Netflix's content strategy.
- Although television content has grown significantly in recent years, movies remain the platform's primary content category.

4.3 Geographic Distribution of Content

Country-level analysis was performed to identify the largest contributors to Netflix's content library.

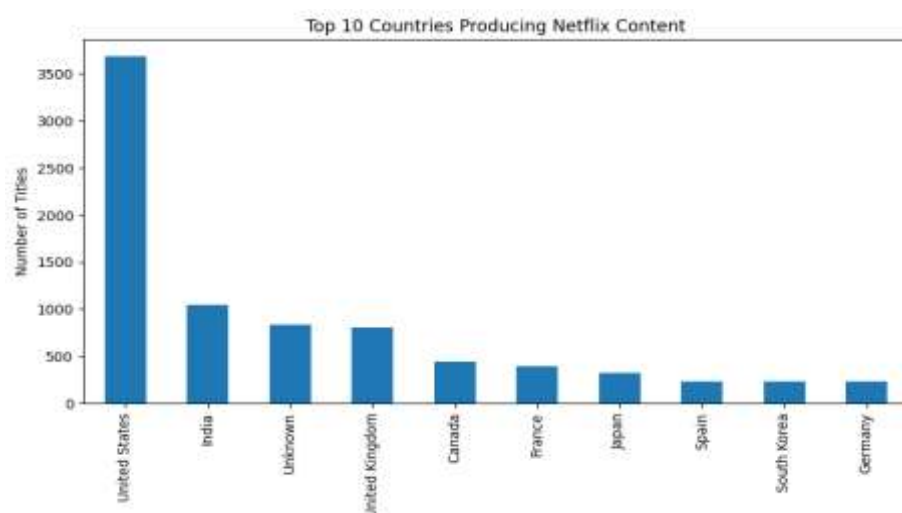


Figure 2: Top 10 Countries Producing Netflix Content

Key observations:

- The United States is the largest content producer, contributing significantly more titles than any other country.
- India ranks second, highlighting Netflix's investment in regional and international markets.
- The United Kingdom, Canada, France, and Japan also contribute substantial volumes of content.
- While Netflix operates globally, content production remains heavily concentrated in a small number of countries.

4.4 Audience Rating Analysis

Content ratings were analyzed to understand the primary audience segments targeted by Netflix.

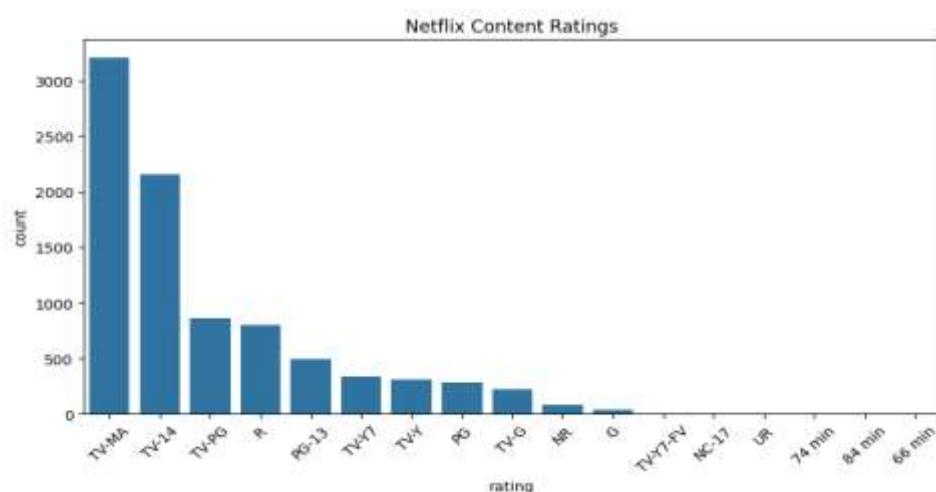


Figure 3: Netflix Content Ratings Distribution

Key observations:

- TV-MA is the most common rating category within the dataset.
- TV-14 is the second most frequent rating.
- The dominance of TV-MA and TV-14 indicates that Netflix primarily targets teenage and adult audiences.

- Family-oriented and children's content is present but represents a smaller proportion of the overall catalogue.

4.5 Genre Analysis

Genre categories were evaluated to identify the most common content types available on Netflix.

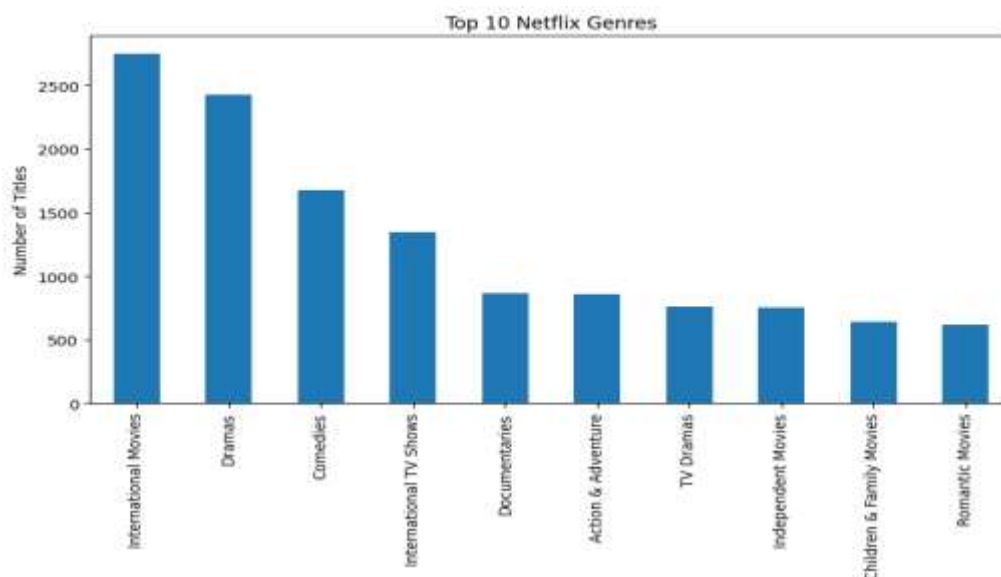


Figure 4: Top 10 Netflix Genres

Key observations:

- International Movies represent the most common genre category.
- Dramas and Comedies are also highly represented across the platform.
- International TV Shows contribute significantly to catalogue diversity.
- The results highlight Netflix's emphasis on global content and broad audience appeal.

4.6 Content Added to Netflix Over Time

The number of titles added to Netflix each year was analyzed to understand platform expansion and content acquisition trends.

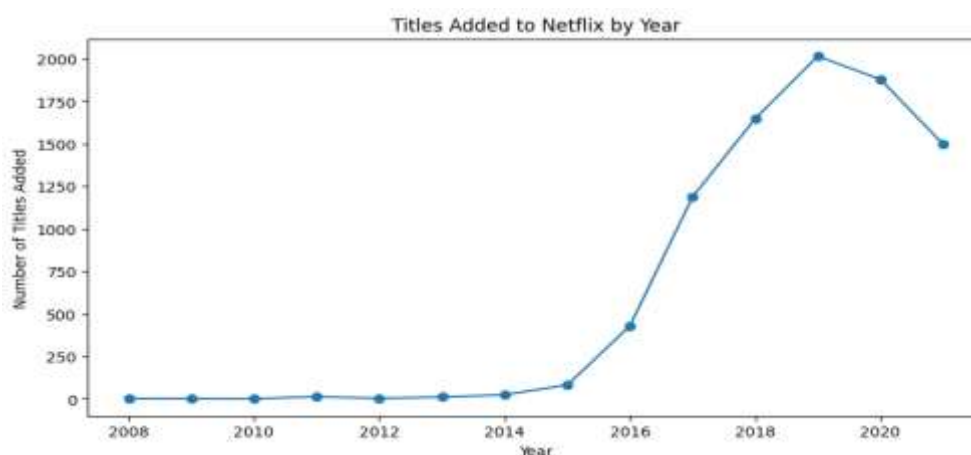


Figure 5: Titles Added to Netflix by Year

Key observations:

- Content additions remained relatively modest until 2015.
- Netflix experienced rapid growth between 2016 and 2019.
- The highest number of content additions occurred in 2019.
- This period reflects Netflix's aggressive content acquisition and global expansion strategy.

4.7 Content Release Trend Analysis

The release years of titles were examined to understand how Netflix's catalogue is distributed across different production periods.

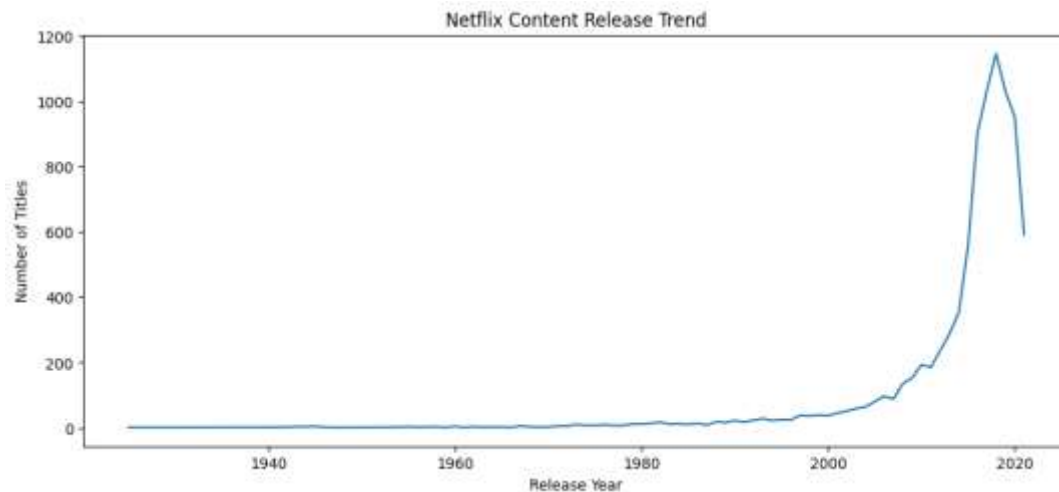


Figure 6: Netflix Content Release Trend

Key observations:

- The majority of Netflix content was released after the year 2000.
- Content production accelerated significantly after 2010.
- Netflix maintains a catalogue that combines both classic and modern productions.
- While older titles remain available, the platform is largely focused on contemporary content.

Overall, the exploratory analysis revealed that Netflix's catalogue is predominantly movie-focused, strongly influenced by content from the United States, largely targeted toward mature audiences, and characterized by substantial growth during the period between 2016 and 2019.

SECTION 05

SQL validation

After the Python analysis, the cleaned dataset was loaded into a PostgreSQL database. Twelve queries were written to verify every key number from a completely separate angle. If Python said 8,797 SQL had to agree.

	Q U E R Y	W H A T I T F O U N D
01	Total records	8,797 titles confirmed matches cleaned dataset
02	Movies vs TV Shows	6,131 movies · 2,666 TV shows
03	Top countries by title count	USA 2,812 · India 972 · UK 418
04	Most common ratings	TV-MA leads with 3,209 titles
05	Titles per release year	Growth visible from 2015 onward
06	Top genres	Dramas, Documentaries, Stand-Up Comedy
07	Oldest title on Netflix	Pioneers: First Women Filmmakers (1925)
08	Most recent titles	Multiple titles from 2021
09	Total movie count	6,131 movies
10	Total TV show count	2,666 TV shows
11	Movie rating breakdown	TV-MA most common for movies too
12	Content added per year	2019 was the single biggest year: 2,016 titles

Every result matched. That cross-check is what gives the numbers in this report their credibility.

SECTION 06

Key Findings

- | | | |
|----|--------------------------------|---|
| 01 | Movies dominate. | 70% of the library is film. Netflix began as a movie platform and that still shapes what you find there. |
| 02 | The USA leads by far. | 2,812 American titles — nearly 3× India's 972. Global yes, but the accent is still American. |
| 03 | It is built for adults. | TV-MA is the most common rating. The platform skews toward mature content by design. |
| 04 | 2019 was the peak year. | 2,016 titles added in a single year. Netflix was at its most aggressive before streaming wars intensified. |
| 05 | Drama is king. | Dramas and International Movies top every genre list. Documentaries and Stand-Up are surprisingly close behind. |
| 06 | The catalogue is old. | Content dating back to 1925 lives on the platform. Netflix is an archive as much as it is a streaming service. |

SECTION 06**Recommendations, Conclusion & Appendix**

Recommendations

- Continue investing in international content to reach global audiences.
- Expand family-friendly content to diversify the catalogue.
- Maintain a balanced mix of movies and TV shows.
- Focus on popular genres such as Drama and International content.
- Monitor content growth trends to support future planning.

Conclusion

This project involved cleaning, analyzing, and validating the Netflix dataset using Python and PostgreSQL. The analysis showed that Netflix's catalogue is movie-focused, dominated by U.S. content, targeted mainly at mature audiences, and experienced significant growth between 2016 and 2019. These findings provide valuable insights into Netflix's content strategy and audience focus.

Appendix**Dataset**

- Netflix Movies and TV Shows Dataset (Kaggle)
- 8,797 records analyzed

Tools Used

- Python (Pandas, NumPy)
- Matplotlib & Seaborn
- PostgreSQL
- Jupyter Notebook

Visualizations

- Movies vs TV Shows
- Top Countries
- Ratings Distribution
- Top Genres
- Content Added by Year
- Release Trends

Deliverables

- Cleaned Dataset (netflix_cleaned.csv)
- Jupyter Notebook
- SQL Queries
- Final Report
- Insight Report